

Green roof heat transfer coefficient measurement and impact of plant species and moisture

Nan Chen, Qianlin Deng, Qiuyi Chen, Zhaolong Wang^{*}

School of Agriculture & Biology, Shanghai Jiao Tong University, Shanghai 200240, PR China

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ABSTRACT

Green roofs attract wide attention for building energy conservation and sustainable development. However, the heat transfer coefficient of green roofs is not well defined. In this study, we developed a non-contact steady-state method for green roof heat transfer coefficient measurement. The apparatus required 16 min to reach the steady state, with a measurement variation of 2.84 ~ 3.47 %. The impacts of plant species and substrate moisture on the heat transfer coefficient of green roofs were defined. The green roof with *Sedum lineare* (a CAM plant) performed the widest variation of heat transfer coefficient, from 2.847 W/m² K (at the wilting point) to 5.401 W/m² K (at the field capacity). The green roof with *Paspalum vaginatum* (a C4 plant) performed the least variation of heat transfer coefficient, with only 2.807 ~ 3.069 W/m² K from the wilting point to the field capacity. The green roof with *Festuca arundinacea* showed mediate variation in heat transfer coefficient, with 2.849 ~ 4.372 W/m² K. This work provides a method to determine green roof heat transfer coefficient and defined dynamics of heat transfer coefficients in three different green roofs, which is crucial to further elucidate the green roof heat transfer mechanism and the impact factors.

1. Introduction

The building is the biggest single contributor to world energy consumption and makes up 40 % of the total energy consumption [4,46]. Building energy conservation is one of the most important strategies for sustainable development and carbon neutrality [25,58]. Green roofs attract wide attention due to their multiple ecological, economic, and social benefits [38,26,54]. Numerous studies indicate that green roofs are nature-based solutions to building energy conservation and improvement of the built environment [3,33,44,66]. Green roofs offer heat insulation via solar radiation absorption and carbon dioxide fixation of plant photosynthesis [21,49], canopy evapotranspiration cooling [13,15,59], and shade-effect of vegetation biomass and their growth substrates [11,22,56]. Tariku and Hagos [53] reported that evaporative cooling from wet growing media and plant transpiration played a significant role in reducing buildings' heat gain. Kostadinovic et al. [27] indicated that the green roof reduced heat transfer by 57 %. Karteris et al. [24] reported that green roofs reduced building heating and cooling consumption by 5 ~ 16 %. Ascione et al. [7] reported that green roofs with tall graminaceous vegetation decreased summer cooling demand by 8.5–10.8 % when compared to traditional roofs in Rome and

Amsterdam. La Roche and Berardi [30] found that the green roof achieved the lowest indoor temperature in summer and the highest indoor temperature in winter. Tang and Zheng [51] found that the green roof reduced the inner surface temperature by 3.2 °C on a summer day. Although green roofs showed an energy-saving effect, the exact heat transfer coefficient of green roofs is not well-defined [50].

The heat transfer coefficient is one of the main evaluation parameters for heat insulation in building envelopes and serves as a general guideline for energy conservation of both industrial and civil buildings [34,60]. The heat transfer coefficients of most building materials are well-defined. The current methods for heat transfer coefficient measurement include contact steady-state methods and non-contact transient methods [6,8,37]. Steady-state methods require that the test material be sandwiched between the hot plate and the cold plate to measure the heat transfer from the hot side to the cold side [8]. The contact resistance between the test material and two plates is a critical concern for the accurate measurement of steady-state methods. It cannot be used for green roofs because the green roof is not stable-sharped and the hot plate will burn the plants. Non-contact transient methods are diffusivity measures of heat transfer through the test materials and the accuracy of measurement is dependent upon the uniformity of the tested

^{*} Corresponding author.

E-mail address: turf@sjtu.edu.cn (Z. Wang).

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materials (Abad et al., 2017). It also cannot be used for the green roof because the green roof is non-uniform with a combination of living plants and their growth substrates. A method that can be used to determine the green roof heat transfer coefficient is urgently needed.

A definition of green roof heat transfer coefficient is an indispensable prerequisite for architects to estimate the green roof performance in building energy conservation. Therefore, the objectives of this study are to 1) establish an easy method for the determination of green roof heat transfer coefficient; 2) understand the dynamics of green roof heat transfer coefficients; 3) define the impacts of plant species and substrate moisture on green roof heat transfer coefficients.

2. Materials and methods

2.1. Methodology

Heat transfer coefficient refers to the heat transferred through an area of 1 square meter with a temperature difference of 1 degree under stable heat transfer conditions. The method is designed based on one-dimensional heat transfer under the steady-state condition. The contact steady-state method (Fig. 1A) is widely used for heat transfer measurement of building materials [6,8,37]. The test material must be hard-shaped and without any deformation when sandwiched between the hot and the cold plates [8]. The loose structure and living vegetation of green roofs make it impossible to use this contact measurement method. Therefore, we developed a non-contact steady-state measurement for green roofs (Fig. 1B). The heater is set up on top of the measurement chamber and the testing green roof module is placed on the bottom floor of the chamber. The downward flow of hot air first warms the canopy of the green roof, then the heat continues downward transfers within the green roof system, and finally reaches the bottom of the green roof. The canopy temperature is recorded as T1 and the green roof bottom temperature is recorded as T2. The thermal meters are placed in the middle of the green roof module to avoid the impact of horizontal heat flow. When heat flow reaches the steady state, the amount of heat transfer can be determined by the temperature gradient as the following equation:

$$Q = k \times A \times (T1 - T2) \quad (1)$$

Where: Q: The amount of heat transferred, W; k: Heat transfer coefficient, W/m² K; A: The area of the heating surface, m²; T1: The temperature of the warming side, °C; T2: The temperature of the cool side, °C.

The green roof is a complex system, including plants, growth substrates, as well as moisture variations. Here, the heat transfer coefficient of a green roof is the equivalent heat transfer coefficient of this complex system based on a proportional relationship between heat transfer and

the temperature gradient. The green roof heat transfer coefficient is measured by comparison with the reference water in the same measurement. The water is selected as the reference because it has the same evaporation property as the green roof and a stable heat transfer coefficient. During the measurement, a green roof module and a reference water (tap water) module are placed in the same measurement chamber to avoid heat conduction errors during heat transfer coefficient measurement. The green roof module is in the same area (A) and thickness (h) as the reference water module (Fig. 1B). The green roof module and the water module received the same amount of heat transferred, therefore, the heat transfer coefficient of the green roof can be calculated by the reference water module.

$$Q_{\text{green roof}} = Q_{\text{water}} \quad (2)$$

That is:

$$k_{\text{green roof}} \times A_{\text{green roof}} \times (T1 - T2) = k_{\text{water}} \times A_{\text{water}} \times (T1' - T2')$$

Because $A_{\text{water}} = A_{\text{green roof}}$,

$$k_{\text{green roof}} \times (T1 - T2) = k_{\text{water}} \times (T1' - T2')$$

The heat transfer coefficient of the green roof can be calculated by the temperature difference between the green roof and the reference water and the known heat transfer coefficient of the reference water, as shown in the following equation:

$$k_{\text{green roof}} = k_{\text{water}} \times (T1' - T2') / (T1 - T2) \quad (3)$$

3. Experiment

3.1. Apparatus and measurement

The heater (HFY20Y, 1200 W, Midea, China) was set up on top of the measurement chamber which was 60 cm above the green roof and the reference water modules. The green roof was grown in the lysimeter (690 mm × 360 mm × 50 mm) with a 5 cm thick peat moss (Pindstrup, Denmark). The extensive green roof module was established with a crassulacean acid metabolism plant (*Sedum lineare*). Green roofs were maintained to their maturity with 100 % canopy coverage and 10 cm of height in the greenhouse.

The measurement was carried out in laboratory conditions, with the air temperature at 22 ± 3 °C and the air relative humidity between 35 ~ 60 %. After adapting to the room temperature for 3 days, the green roof module and a reference water module of the same size were placed on the floor of the chamber for heat transfer coefficient measurement. When the measurement started, the chamber was closed to avoid heat escape. Thermal meters were set to record the canopy temperature (T1) and bottom temperature (T2) of the green roof and surface temperature

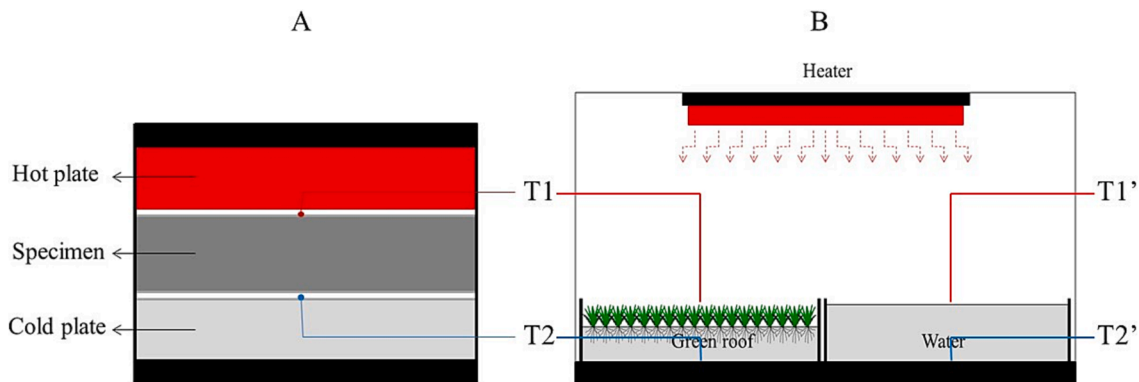


Fig. 1. Schematic diagram of steady-state methods for heat transfer coefficient measurement. A: The contact heat transfer method. B: Non-contact Steady-state method for green roofs. T1 and T2: the temperatures of the warm side and cool side in the green roof module. T1' and T2': the temperatures of the warm side and cool side in the reference water module.

(T1') and bottom temperature (T2') of the reference water module as shown in Fig. 2. Each temperature was recorded in a 1-minute interval.

3.2. Evaluation

To evaluate the accuracy and feasibility of the non-contact steady-state method in this study, a sample of deionized water at 21°C was measured by this non-contact steady-state method, and a heat transfer instrument of transient laser flash method (LFA 467, Netzsch, Germany) in the room temperature for the comparison of the measurement deviation. All measurements were repeated four times.

3.3. Heat transfer coefficient of green roofs

Green roofs with 3 plant species (*Sedum lineare*, a crassulacean acid metabolism plant species; *Festuca arundinacea*, a C3-plant species; and *Paspalum vaginatum*, a C4-plant species) were used for their heat transfer coefficient measurement. The experiment was conducted in spring and all 3 plant species are in their normal growth conditions. Green roofs were fully watered to reach their field capacity (no excess water flows from the drainage holes at the bottom of the lysimeter). After saturation, all green roofs were in the greenhouse for their natural evapotranspiration to form their different substrate moisture gradients. The moisture of growth substrates was determined every day by the weighting method [32]. Green roof modules with different substrate moisture were measured by the non-contact steady-state method. The measurement for each plant species with the different substrate moisture gradients was repeated at least 3 times.

4. Results and discussions

4.1. Steady state for heat transfer coefficient measurement

After turning on the heater, the heat flow downward, and the surface temperature of the green roof canopy (T1) and water surface (T1') increased sharply, reaching the highest platform after 23 and 25 min of heating, respectively (Fig. 2A, B). However, the bottom temperature showed a slower increase when compared to the surface temperature, both in the green roof module (T2) and the reference water module (T2'). The bottom temperature of the green roof (T2) and the reference water (T2') reached their stable platform after 28 and 33 min of heating, respectively. The temperature difference between the surface and bottom of the green roof and the reference water module is shown in Fig. 2 C, D. Both of them reached the stable platform after 16 min of heating, indicating that the heat transfer in this apparatus could reach a steady state after 16 min.

By comparing the temperature difference between the green roof module (T1-T2) and the reference water module (T1'-T2'), the heat transfer coefficient of the green roof can be calculated with Equation (3). Fig. 3 shows the stabilization times required to reach the steady state for a green roof heat transfer coefficient measurement. Heat flow was stabilized after 16 min of heating in the measurement of this apparatus. The heat transfer coefficient of the test green roof was maintained consistently at $4.705 \pm 0.012 \text{ W/m}^2 \text{ K}$, with a 1.35 % variation during the steady state period.

Steady-state heat flux is a standard test method for heat transfer measurement [8]. The principle of this standard method is shown in Fig. 1A. The flat specimen is placed in contact with two copper plates (a

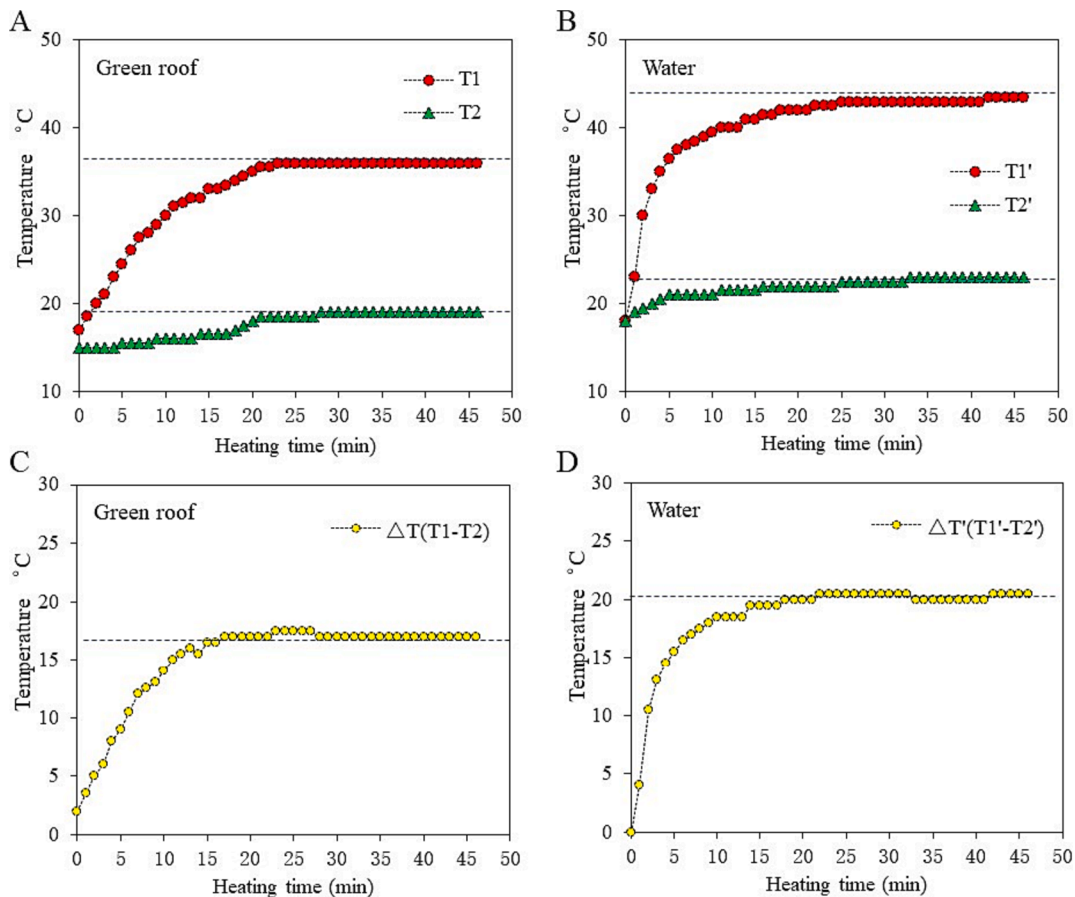


Fig. 2. Temperature dynamics of the heating side (T1) and cool side (T2) in the green roof module and the water module (T1' and T2'). A: Temperature dynamic of T1 and T2 in the green roof module. B: Temperature dynamic of T1' and T2' in the water module. C: Temperature difference ΔT (T1-T2) in the green roof module. D: Temperature difference $\Delta T'$ (T1'-T2') in the water module.

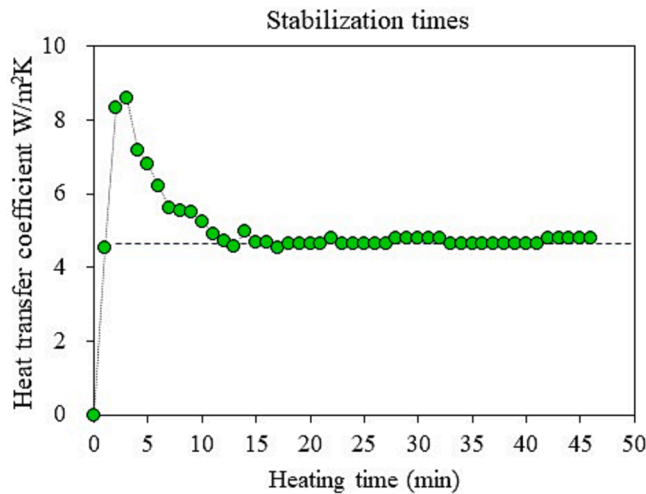


Fig. 3. Stabilization times for the heat transfer coefficient measurement in a green roof with *Sedum lineare* and 83.5% of substrate moisture (field capacity, saturated).

hot plate on one side and a cold plate on the other side). The heat transfer coefficient is calculated by the constant heat flow and the temperature difference between the specimen surfaces (T1 and T2). The temperature of the hot plate can be set differently according to the properties of the test materials [61]. In oil palm shell foamed concrete with conventional materials, Alengaram et al. (2013) set temperatures of the hot plate at 40 °C and the cold plate at 18 °C. The heat transfer measured by the steady-state method showed a variation in the temperature setting in the cement concrete [18]. The steady-state method requires a long period to reach the steady heat flux throughout the test material from the hot plate to the cold plate for heat transfer measurement. Most studies used a 24-hour period in the measurement [8,35]. Some studies even extended the measurement hours to 48 h to ensure

more consistent and accurate measurements [2]. In this study, non-contact steady-state requires only 16 min to get the steady state and accurate data. It provides a quick and efficient measurement of heat transfer coefficient.

4.2. Evaluation with laser flash method

The accuracy of the measurement is an important consideration for the feasibility of the method. Here we compared the non-contact steady-state method and a transient laser flash method by the instrument FA 467 (Netzsch, Germany) for the measurement evaluation. The heat transfer coefficient of the deionized water by the Laser flash method was 3.813 ± 0.041 W/m² K, with a 2.15 % variation. The non-contact steady-state method from 15 to 30 min after heating also resulted in consistent heat transfer coefficient measurement, with 3.811 ± 0.116 W/m² K, and the coefficient of variation was between 2.62 ~ 3.47 % (Fig. 4). No significant difference was noticed in the value of heat transfer coefficient measured by these two methods. Compared to the laser flash instrument, the non-contact steady-state method only showed slightly higher standard errors and coefficient of variation during the measurements.

It is normal for variations in the heat transfer measurement. A precise measurement with a 0.7 % variation was reported by Malinaric [37] in a step-wise transient method for solid materials. Sabapathy & Gedupudi [47] reported a 3.87 % repeatability deviation in the measurement of straw bale samples. The variation reached 5 ~ 12.5 % in the measurement of the cut tobacco by the transient plane source method [65]. The greater deviations in heat transfer coefficient measurement occurred when using the prediction model calculations [62]. Pianella et al. [40] compared steady-state and transient methods in thermal conductivities of growth substrates and found that steady-state measurements showed more consistency with a variation of 4 ~ 7 % for dry and wet samples. In this study, the coefficient of variations of heat transfer coefficient measurements was 2.84 ~ 3.47 % by the non-contact steady-state method, which is acceptable when compared to the previous studies.

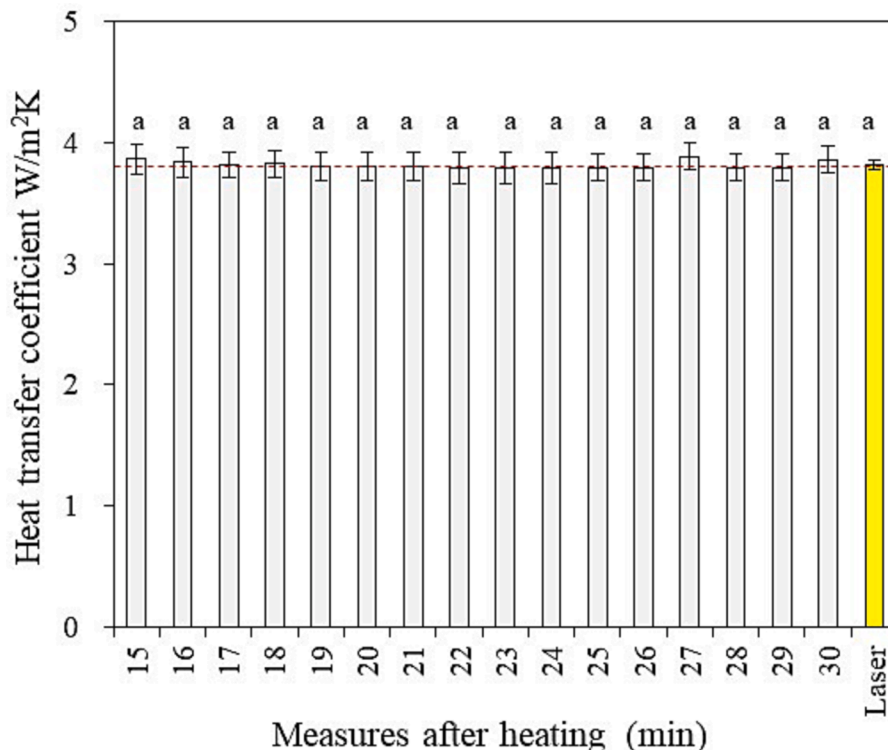


Fig. 4. Comparison of heat transfer coefficient measurements by Steady-state and Laser flash (Netzsch, LFA 467).

4.3. Green roof heat transfer coefficient dynamics

The heat transfer properties of building envelopes attract wide attention in building energy conservation [43]. Most building materials have been well-studied on their heat transfer performance. The heat transfer coefficients of brick and concrete walls with 40 cm thickness were 1.75 ~ 2.17 W/m² K [29]. The heat transfer coefficient of the clay and concrete bricks was 5.43 W/m² K, while the fly ash brick could lower to 2.42 W/m² K [31]. Venkatesan et al. [55] reported that heat transfer coefficients of red clay brick walls were very close to the standard value of 5.5 W/m² K with the maximum deviation was 4.5 %.

However, there is little research on the heat transfer coefficient of green roofs. The value of the heat transfer coefficient of a general green roof is still unknown, which limits the implementation of green roofs in building energy conservation [63,64]. It is difficult to quantify the heat transfer coefficient of a living green roof. Therefore, many earlier studies focused on the measurement of green roof growth substrates because they can be measured by the existing instruments [19,40,48]. Growth substrates are completely different from green roofs. Many studies indicated that plants may play a more important role than their substrates in green roof heat insulation [14,63]. Limited data only come from the comparison of the heat transfer difference between the green roof and the bare roof during the summer months [20,51,64]. Collins et al. [16] reported the thermal behavior of heating loss during the winter by simulating a green roof module and a 25 W light bulb heating room below it.

In this study, a non-contact steady-state method was developed to directly measure the heat transfer coefficient of green roofs. The heat transfer coefficient of each sample was calculated as the average of 25 ~ 30 min after heating according to the stabilization time of this method (Fig. 3). The results showed that green roofs with 3 different plant species performed different dynamic ranges of heat transfer coefficient (Table 1). All green roofs showed the minimum heat transfer coefficients at substrate moisture at the wilting point and the maximum heat transfer coefficients at substrate moisture at the field capacity. The green roof with *Sedum lineare* showed the widest variation of heat transfer coefficient, ranging from 2.847 W/m² K (wilting point) to 5.401 W/m² K (field capacity), followed by the green roof with *Festuca arundinacea*, ranging from 2.849 W/m² K (wilting point) to 4.372 W/m² K (field capacity). The green roof with *Paspalum vaginatum* showed the least variation of heat transfer coefficient, only ranging from 2.807 W/m² K (wilting point) to 3.069 W/m² K (field capacity).

4.4. Heat transfer coefficient affected by moisture

Numerous research indicates that the moisture content significantly affects the heat transfer of building insulation materials [9,52,60]. The measurement of heat transfer coefficient with changes in moisture content is vitally important for the accurate calculation of the energy

conservation of the building envelope [1]. Some researchers found that there is a linear relationship between heat transfer and water content in fibrous insulation materials [1], tropical wood species [10], and aerated concrete [23]. However, Belkharchouche and Chaker [9] indicated that the heat transfer of the lightened construction material was exponentially increased with its water content.

The moisture of the green roof presents a dynamic change with a combination of rainfall, irrigation, and evapotranspiration [59]. Runoff occurs after the rainfall exceeds the maximum retention capacity of the green roof at field capacity [32]. Irrigation must be applied when plants show wilting symptoms at the wilting point to maintain plant survival, otherwise, plants will be desiccated to die. Therefore, the field capacity and wilting point represent the maximum and minimum moisture of the living green roof [57]. The moisture dynamic of the green roof could also result in the dynamic of its heat transfer coefficient. Therefore, the dynamics of the heat transfer coefficient of green roofs with 3 different plant species were evaluated in this study.

The heat transfer coefficient of green roofs with all three plant species showed a significant increase with the increase of substrate moisture. However, the dynamic difference among these three plant species is noticed (Fig. 5). The green roof with *Sedum lineare* (a CAM plant) showed its wilting point when the substrate moisture dropped to 53.1 %, and the field capacity when the substrate moisture reached 82.9 %. The heat transfer coefficient of the green roof with *Sedum lineare* increased in a quadratic curve with the increase of substrate moisture and fits in the following equation: $y = 0.0014x^2 - 0.1003x + 4.5092$ (Fig. 5A). The green roof with *Festuca arundinacea* (a C3 plant) showed its wilting point when the substrate moisture dropped to 77.5 %, and the field capacity when the substrate moisture reached 85.2 %. The heat transfer coefficient of the green roof with *Festuca arundinacea* showed a linear increase and fits in the following equation: $y = 0.03323x + 1.5988$ (Fig. 5B). The green roof with *Paspalum vaginatum* (a C4 plant) showed its wilting point when the substrate moisture dropped to 76.2 %, and the field capacity when the substrate moisture reached 82.6 %. The heat transfer coefficient of the green roof with *Paspalum vaginatum* increased in a quadratic curve and fits in the following equation: $y = 0.0007x^2 - 0.00736x + 44.733$ (Fig. 5C).

The heat transfer of green roofs was a combination of the canopy layer and their growth substrate integrated with the root system. Almeida et al. [5] studied the heat insulation of the substrate and vegetation layers and found that the vegetation layer was of key importance to the overall performance of the green roof. The heat insulation of green roofs may be attributed to water evapotranspiration, radiation absorption, photosynthetic assimilation, and solar shading of the plant canopy [13]. In this study, three plant species used in green roofs were in different photosynthetic pathways in carbon assimilation. *Sedum lineare* is a crassulacean acid metabolism plant species [28], *Festuca arundinacea* is a C3 plant species [17], and *Paspalum vaginatum* is a C4 plant species [36]. There was a significant difference in transpiration cooling among these plant species [13]. Here, the different heat transfer coefficients of green roofs with these three plant species in response to substrate moisture indicate that plant metabolism might play a key regulatory role in the heat transfer of green roofs.

It should be noticed that the heat transfer coefficients of these three green roofs are almost at the same level at their wilting points, that is 2.847 W/m² K in the green roof with *Sedum lineare*, 2.849 W/m² K in the green roof with *Festuca arundinacea*, and 2.807 W/m² K in the green roof with *Paspalum vaginatum* (Table 1), ignoring the great difference among their substrate moistures (Fig. 5, 53.1 % in the green roof with *Sedum lineare*, 77.5 % in the green roof with *Festuca arundinacea*, and 76.2 % in the green roof with *Paspalum vaginatum*, respectively). The similar heat transfer coefficients among the different green roofs at the wilting point could be explained by the physiologically inactive of the wilting plants. On the contrary, the heat transfer coefficients of green roofs at their field capacities showed significant differences (Table 1, 5.401 W/m² K in the green roof with *Sedum lineare*, 4.372 W/m² K in the green roof with

Table 1

Heat transfer coefficient variation of green roofs with different plant species.

Green roof types	Heat transfer coefficient variation W/m ² K		Measures	Coefficient of variation (%)
	Minimum (Wilting point)	Maximum (Field capacity)		
<i>Sedum lineare</i>	2.847 ± 0.013 a	5.401 ± 0.027 a	8	0.84 ~ 1.01
<i>Festuca arundinacea</i>	2.849 ± 0.033 a	4.372 ± 0.033b	8	1.57 ~ 2.36
<i>Paspalum vaginatum</i>	2.807 ± 0.006 a	3.069 ± 0.027c	8	0.35 ~ 1.63

The different letters (a, b, c) indicate statistical significance among the green roof types (p < 0.05).

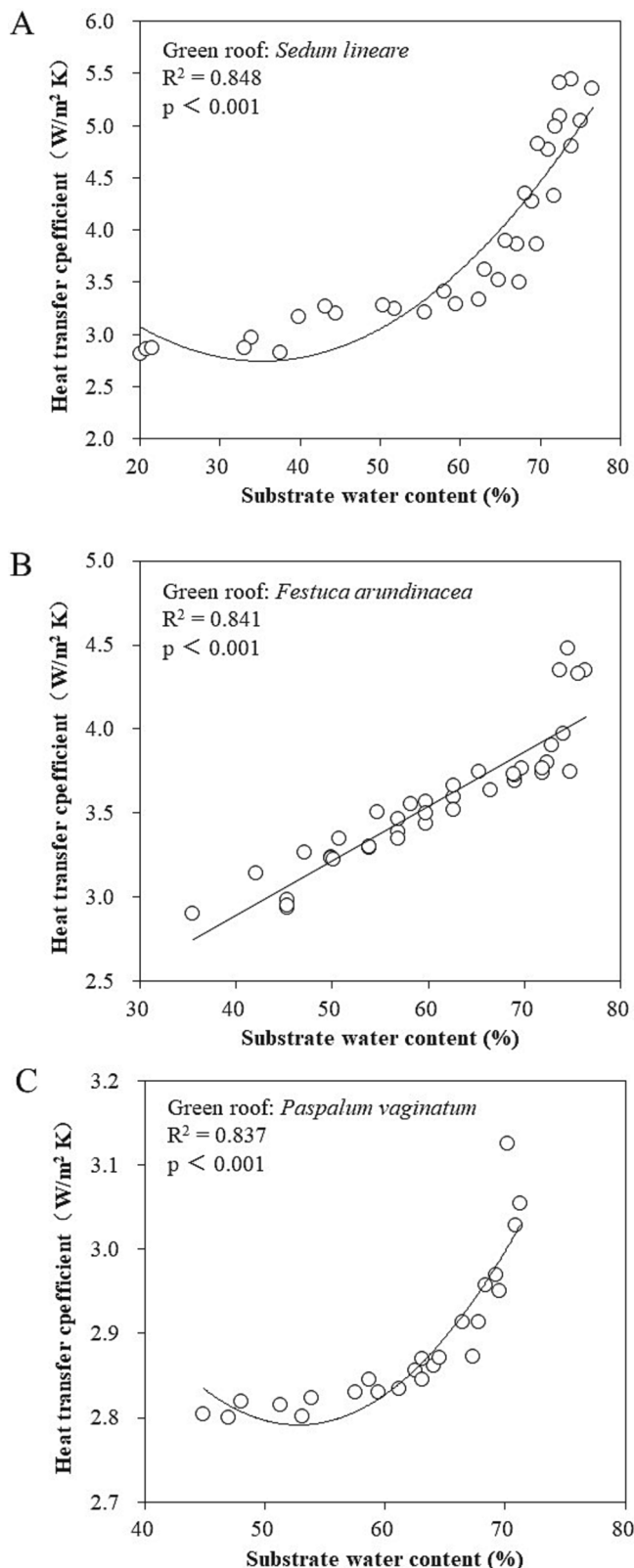


Fig. 5. Heat transfer coefficient of green roofs in response to the substrate moisture. A: green roof with *Sedum lineare*; B: green roof with *Festuca arundinacea*; C: green roof with *Paspalum vaginatum*. The green lines represent the 95% confidence interval.

Festuca arundinacea, and $3.069 \text{ W/m}^2 \text{ K}$ in the green roof with *Paspalum vaginatum*), although there only minor difference of their substrate moistures among them (Fig. 5, 82.9 % in the green roof with *Sedum lineare*, 85.2 % in the green roof with *Festuca arundinacea*, and 82.6 % in the green roof with *Paspalum vaginatum*). Plant species with different photosynthetic pathways showed a significant impact on the heat transfer coefficients of green roofs (Table 1 and Fig. 5). Our results indicate that plant physiological metabolisms played a critical role in the heat transfer performance of green roofs rather than the substrate layer in the green roof system.

The heat transfer performance of green roofs may be impacted by various vegetation types [50], their growth habits [13], canopy density [12,39], substrate composition [11,42], and depth [41,45].

This research provided a simple method for green roof heat transfer coefficient measurement and defined the dynamics of the heat transfer coefficients of 3 plant species with different pathways in light energy assimilations. The results of the heat transfer coefficient dynamics of green roofs can be referred to in the calculation of green roof energy-saving in architectural design.

This study was conducted in optimal laboratory conditions. However, the real-time climate factors such as sunlight, temperature, moisture, and wind of outdoor green roofs are changing, and the plant growth also varies with the different seasons. All these will affect the green roof heat transfer.

Based on this method, further detailed research could be conducted to study the heat transfer performance of green roofs, which will help to elucidate the green roof heat transfer mechanism and its impact factors. An accurate prediction model could be expected to be established for estimating the green roof contributions to building energy conservation after a well-defining heat transfer behavior of green roofs, including vegetation types, their seasonal changes, energy assimilation characteristics, climate changes, substrate components, depth, and moisture dynamics. Therefore, architects could estimate the green roof contribution to the overall building energy conservation based on the data of heat transfer coefficient measurement.

5. Conclusions

This study presented a non-contact steady-state method to determine the heat transfer coefficient of green roofs. This apparatus requires 16 min to reach the steady state of heat transfer for the green roof measurement. The measurement is accurate and reliable with a 2.84 ~ 3.47 % variation. The heat transfer coefficients of green roofs with three different plant species (*Sedum lineare*, *Festuca arundinacea*, and *Paspalum vaginatum*) were defined. The green roof with *Sedum lineare* (CAM plant) showed the widest range of heat transfer coefficient, followed by the green roof with *Festuca arundinacea* (C3 plant). Green roof with *Paspalum vaginatum* (C4 plant) showed the least range of heat transfer coefficient. These different plant species showed different dynamics of heat transfer coefficient in response to substrate moisture, indicating that plants play a critical role in the heat transfer of green roofs. This work provides an easy method to determine the heat transfer coefficient of green roofs and the possibility of further elucidating the green roof heat transfer mechanism and its impact factors.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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References

- [1] A. Abdou, I. Budaiwi, The variation of thermal conductivity of fibrous insulation materials under different levels of moisture content, *Constr Build Mater* 43 (2013) 533–544, <https://doi.org/10.1016/j.conbuildmat.2013.02.058>.
- [2] U.J. Alengaram, B.A. Al Muhit, bin, M.Z. Jumaat, M.L.Y. Jing, A comparison of the thermal conductivity of oil palm shell foamed concrete with conventional materials, *Mater Des* 51 (2013) 522–529, <https://doi.org/10.1016/j.matdes.2013.04.078>.
- [3] M.A. Alim, A. Rahman, Z. Tao, B. Garner, R. Griffith, M. Liebman, Green roof as an effective tool for sustainable urban development: An Australian perspective in relation to stormwater and building energy management, *J Clean Prod* 362 (2022), 132561, <https://doi.org/10.1016/j.jclepro.2022.132561>.
- [4] A. Allouhi, Y. El Fouih, T. Kouskou, A. Jamil, Y. Zeraoui, Y. Mourad, Energy consumption and efficiency in buildings: current status and future trends, *J Clean Prod* 109 (2015) 118–130, <https://doi.org/10.1016/j.jclepro.2015.05.139>.
- [5] R. Almeida, N. Simoes, A. Tadeu, P. Palha, J. Almeida, Thermal behaviour of a green roof containing insulation cork board. An experimental characterization using a bioclimatic chamber, *Build Environ* 160 (2019), 106179, <https://doi.org/10.1016/j.buildenv.2019.106179>.
- [6] I. Asadi, P. Shafagh, Z.F. Bin Abu Hassan, N.B. Mahyuddin, Thermal conductivity of concrete-a review, *J Build Eng* 20 (2018) 81–93, <https://doi.org/10.1016/j.job.2018.07.002>.
- [7] F. Ascione, N. Bianco, F. de' Rossi, G. Turni, G.P. Vanoli, Green roofs in European climates. Are effective solutions for the energy savings in air-conditioning? *Appl Energy* 104 (2013) 845–859, <https://doi.org/10.1016/j.apenergy.2012.11.068>.
- [8] ASTM c177, Standard test method for steady-state heat flux measurements and thermal transmission properties by means of the guarded-hot-plate, *ASTM Int* (2013) 1–23, <https://doi.org/10.1520/C0177-13.2>.
- [9] D. Belkharouch, A. Chaker, Effects of moisture on thermal conductivity of the lightened construction material, *Int J Hydrog Energy* 41 (2016) 7119–7125, <https://doi.org/10.1016/j.ijhydene.2016.01.160>.
- [10] F. Bobda, J.C. Damfeu, R.R.N. Mvondo, P. Meukam, Y. Jannot, Thermal properties measurement of two tropical wood species as a function of their water content using the parallel hot wire method, *Constr Build Mater* 320 (2022), 125974, <https://doi.org/10.1016/j.conbuildmat.2021.125974>.
- [11] M.A. Bollman, G.E. DeSantis, R.S. Waschmann, P.M. Mayer, Effects of shading and composition on green roof media temperature and moisture, *J Environ Manage* 281 (2021), 111882, <https://doi.org/10.1016/j.jenvman.2020.111882>.
- [12] M. Buckland-Nicks, A. Heim, J. Lundholm, Spatial environmental heterogeneity affects plant growth and thermal performance on a green roof, *Sci Total Environ* 553 (2016) 20–31, <https://doi.org/10.1016/j.scitotenv.2016.02.063>.
- [13] J.J. Cao, S. Hu, Q. Dong, L.J. Liu, Z.L. Wang, Green roof cooling contributed by plant species with different photosynthetic strategies, *Energy Build* 195 (2019) 45–50, <https://doi.org/10.1016/j.enbuild.2019.04.046>.
- [14] S. Cascone, A. Gagliano, T. Poll, G. Sciuto, Thermal performance assessment of extensive green roofs investigating realistic vegetation-substrate configurations, *Build Simul* 12 (2019) 379–393, <https://doi.org/10.1007/s12273-018-0488-y>.
- [15] P.Y. Chen, Effects of meteorological variables and substrate moisture on evapotranspiration and thermal performance of a green roof in a subtropical climate, *Ecol Eng* 180 (2022), 106663, <https://doi.org/10.1016/j.ecoleng.2022.106663>.
- [16] S. Collins, K. Kuoppamäki, D.J. Kotze, X.S. Lu, Thermal behavior of green roofs under Nordic winter conditions, *Build Environ* 122 (2017) 206–214, <https://doi.org/10.1016/j.buildenv.2017.06.020>.
- [17] T. Culpepper, J. Young, D.T. Montague, D. Sullivan, B. Wherley, Physiological responses in C-3 and C-4 turfgrasses under soil water deficit, *HortSci* 54 (2019) 2249–2256, <https://doi.org/10.21273/HORTSCI14357-19>.
- [18] A.S. Gandage, V.R.V. Rao, M.V.N. Sivakumar, A. Vasan, M. Venu, A.B. Yaswanth, Effect of perlite on thermal conductivity of self compacting concrete, *Procedia – Soc Behav Sci* 104 (2013) 188–197, <https://doi.org/10.1016/j.sbspro.2013.11.111>.
- [19] N. Gerzhova, J. Cote, S. Menard, A Conceptual framework for modelling the thermal conductivity of dry green roof substrates, *Resources* 14 (2019) 8573–8599, <https://doi.org/10.15376/biores.14.4.8573-8599>.
- [20] Y. He, H. Yu, A. Ozaki, N.N. Dong, S.L. Zheng, Long-term thermal performance evaluation of green roof system based on two new indexes: a case study in Shanghai area, *Build Environ* 120 (2017) 13–28, <https://doi.org/10.1016/j.buildenv.2017.04.001>.
- [21] G. Heidarinejad, A. Esmaili, Numerical simulation of the dual effect of green roof thermal performance, *Energy Convers Manage* 106 (2015) 1418–1425, <https://doi.org/10.1016/j.enconman.2015.10.020>.
- [22] C.Y. Jim, Building thermal-insulation effect on ambient and indoor thermal performance of green roofs, *Ecol Eng* 69 (2014) 265–275, <https://doi.org/10.1016/j.ecoleng.2014.04.016>.
- [23] H.Q. Jin, X.L. Yao, L.W. Fan, X. Xu, Z.T. Yu, Experimental determination and fractal modeling of the effective thermal conductivity of autoclaved aerated concrete: effects of moisture content, *Int J Heat Mass Tran* 92 (2016) 589–602, <https://doi.org/10.1016/j.ijheatmasstransfer.2015.08.103>.
- [24] M. Karteris, I. Theodoridou, G. Mallinis, E. Tsiros, A. Karteris, Towards a green sustainable strategy for Mediterranean cities: Assessing the benefits of large-scale green roofs implementation in Thessaloniki, Northern Greece, using environmental modelling, GIS and very high spatial resolution remote sensing data, *Renew Sustain Energy Rev* 58 (2016) 510–525, <https://doi.org/10.1016/j.rser.2015.11.098>.
- [25] H. Kim, H. Choi, H. Kang, J. An, S. Yeom, T. Hong, A systematic review of the smart energy conservation system: from smart homes to sustainable smart cities, *Renew Sustain Energy Rev* 140 (2021), 110755, <https://doi.org/10.1016/j.rser.2021.110755>.
- [26] E. Koroxenidis, T. Theodosiou, Comparative environmental and economic evaluation of green roofs under Mediterranean climate conditions – extensive green roofs a potentially preferable solution, *J Clean Prod* 311 (2021), 127563, <https://doi.org/10.1016/j.jclepro.2021.127563>.
- [27] D. Kostadinovic, M. Jovanovic, V. Bakic, N. Stepanic, M. Todorovic, Experimental investigation of summer thermal performance of the green roof system with mineral wool substrate, *Build Environ* 217 (2022), 109061, <https://doi.org/10.1016/j.buildenv.2022.109061>.
- [28] T. Kurokuma, H. Watanabe, Physiological and morphological traits and competence for carbon sequestration of several green roof plants under a controlled environmental system, *J Am Soc Hort Sci* 141 (2016) 583–590, <https://doi.org/10.21273/JASHS03909-16>.
- [29] A. Lakatos, F. Kalmar, Examination of the change of the overall heat transfer coefficients of building structures in function of water content, *Build Serv Eng Res T* 35 (2014) 507–515.
- [30] P. La Roche, U. Berardi, Comfort and energy savings with active green roofs, *Energy Build* 82 (2014) 492–504, <https://doi.org/10.1016/j.enbuild.2014.07.055>.
- [31] J. Li, W. Cao, G. Chen, The heat transfer coefficient of new construction - brick masonry with fly ash blocks, *Energy* 86 (2015) 240–246, <https://doi.org/10.1016/j.energy.2015.04.028>.
- [32] X.X. Li, J.J. Cao, P.X. Xu, L. Fei, Q. Dong, Z.L. Wang, Green roofs: Effects of plant species used on runoff, *Land Degrad Dev* 29 (2018) 3628–3638, <https://doi.org/10.1002/ldr.3102>.
- [33] M. Lin, J. Dong, L. Jones, J. Liu, T. Lin, J. Zuo, H. Ye, G. Zhang, T. Zhou, Modeling green roofs' cooling effect in high-density urban areas based on law of diminishing marginal utility of the cooling efficiency: a case study of Xiamen Island, China, *J Clean Prod* 316 (2021), 128277, <https://doi.org/10.1016/j.jclepro.2021.128277>.
- [34] L.K. Liu, D. Su, Y.J. Tang, G.Y. Fang, Thermal conductivity enhancement of phase change materials for thermal energy storage: a review, *Renew Sustain Energy Rev* 62 (2016) 305–317, <https://doi.org/10.1016/j.rser.2016.04.057>.
- [35] M.Y.J. Liu, U.J. Alengaram, M.Z. Jumaat, K.H. Mo, Evaluation of thermal conductivity, mechanical and transport properties of lightweight aggregate foamed geopolymer concrete, *Energy Build* 72 (2014) 238–245, <https://doi.org/10.1016/j.enbuild.2013.12.029>.
- [36] F. Lulli, M. Volterrani, N. Grossi, R. Armeni, S. Stefanini, L. Guglielminetti, Physiological and morphological factors influencing wear resistance and recovery in C-3 and C-4 turfgrass species, *Funct Plant Biol* 39 (2012) 214–221, <https://doi.org/10.1071/FP11234>.
- [37] S. Malinaric, Step-wise transient method, *Meas Sci Technol* 27 (2016), 035601, <https://doi.org/10.1088/0957-0233/27/3/035601>.
- [38] M. Manso, I. Teotonio, C.M. Silva, C.O. Cruz, Green roof and green wall benefits and costs: a review of the quantitative evidence, *Renew Sustain Energy Rev* 135 (2021), 110111, <https://doi.org/10.1016/j.rser.2020.110111>.
- [39] G. Perez, A. Vila, C. Sole, J. Coma, A. Castell, L.F. Cabeza, The thermal behaviour of extensive green roofs under low plant coverage conditions, *Energy Effic* 8 (2015) 881–894, <https://doi.org/10.1007/s12053-015-9329-3>.
- [40] A. Pianella, R.E. Clarke, N.S.G. Williams, Z.D. Chen, L. Aye, Steady-state and transient thermal measurements of green roof substrates, *Energy Build* 131 (2016) 123–131, <https://doi.org/10.1016/j.enbuild.2016.09.024>.
- [41] A. Pianella, L. Aye, Z.D. Chen, N.S.G. Williams, Substrate depth, vegetation and irrigation affect green roof thermal performance in a Mediterranean type climate, *Sustainability* 9 (2017) 1451, <https://doi.org/10.3390/su9081451>.
- [42] Rahman MS, MacPherson S, Lefsrud M. Prospects of porous concrete as a plant-growing medium and structural component for green roofs: a review. *Ren Agri Food Sys* 2022; PII S1742170522000138. DOI: 10.1017/S1742170522000138.
- [43] B. Raji, M.J. Tenpierik, A. van den Dobbelsteen, The impact of greening systems on building energy performance: a literature review, *Renew Sustain Energy Rev* 45 (2015) 610–623, <https://doi.org/10.1016/j.rser.2015.02.011>.
- [44] J.D. Ran, M.F. Tang, Passive cooling of the green roofs combined with night-time ventilation and walls insulation in hot and humid regions, *Sustain Cities Soc* 38 (2018) 466–475, <https://doi.org/10.1016/j.scs.2018.01.027>.
- [45] R. Reyes, W. Bustamante, J. Gironas, P.A. Pasten, V. Rojas, F. Suarez, S. Vera, F. Victorero, C.A. Bonilla, Effect of substrate depth and roof layers on green roof temperature and water requirements in a semi-arid climate, *Eco Engineer* 97 (2016) 624–632, <https://doi.org/10.1016/j.ecoleng.2016.10.025>.
- [46] C. Robinson, B. Dilkina, J. Hubbs, W. Zhang, S. Guhathakurta, M.A. Brown, R. M. Pendyala, Machine learning approaches for estimating commercial building energy consumption, *Appl Energy* 208 (2017) 889–904, <https://doi.org/10.1016/j.apenergy.2017.09.060>.
- [47] K.A. Sabapathy, S. Gedupudi, Straw bale based constructions: Measurement of effective thermal transport properties, *Constr Build Mater* 198 (2019) 182–194, <https://doi.org/10.1016/j.conbuildmat.2018.11.256>.

- [48] D.J. Sailor, M. Elagos, An updated and expanded set of thermal property data for green roof growing media, *Energy Build* 43 (2011) 2298–2303, <https://doi.org/10.1016/j.enbuild.2011.05.014>.
- [49] O. Starry, J.D. Lea-Cox, J. Kim, M.W. van Iersel, Photosynthesis and water use by two Sedum species in green roof substrate, *Environ Exp Bot* 107 (2014) 105–112, <https://doi.org/10.1016/j.envexpbot.2014.05.014>.
- [50] T. Susca, Green roofs to reduce building energy use? A review on key structural factors of green roofs and their effects on urban climate, *Build Environ* 162 (2019), 106273, <https://doi.org/10.1016/j.buildenv.2019.106273>.
- [51] M.F. Tang, X. Zheng, Experimental study of the thermal performance of an extensive green roof on sunny summer days, *Appl Energy* 242 (2019) 1010–1021, <https://doi.org/10.1016/j.apenergy.2019.03.153>.
- [52] D. Taoukil, A. El bouardi, F. Sick, A. Mimet, H. Ezbakhe, T. Ajzoul, Moisture content influence on the thermal conductivity and diffusivity of wood-concrete composite, *Constr Build Mater* 48 (2013) 104–115.
- [53] F. Tariku, S. Hagos, Performance of green roof installed on highly insulated roof deck and the plants' effect: an experimental study, *Build Environ* 221 (2022), 109337, <https://doi.org/10.1016/j.buildenv.2022.109337>.
- [54] I. Teotonio, C.M. Silva, C.O. Cruz, Economics of green roofs and green walls: a literature review, *Sustain Cities Soc* 69 (2021), 102781, <https://doi.org/10.1016/j.scs.2021.102781>.
- [55] M. Venkatesan, M. Raja, S. Sivalaksmi, S. Suresh, Experimental study of heat transfer coefficients on red clay brick wall, *J Therm Anal Calorim* 147 (2022) 5949–5959, <https://doi.org/10.1007/s10973-021-10896-y>.
- [56] S. Vera, C. Pinto, P.C. Tabares-Velasco, W. Bustamante, F. Victorero, J. Gironas, C. A. Bonilla, Influence of vegetation, substrate, and thermal insulation of an extensive vegetated roof on the thermal performance of retail stores in semiarid and marine climates, *Energy Build* 146 (2017) 312–321, <https://doi.org/10.1016/j.enbuild.2017.04.037>.
- [57] K. Vijayaraghavan, Green roofs: A critical review on the role of components, benefits, limitations and trends, *Renew Sustain Energy Rew* 57 (2016) 740–752, <https://doi.org/10.1016/j.rser.2015.12.119>.
- [58] K. Wang, Y. Aktas, L. Malki-Epshtein, D. Wu, M.F.A. Bin Abdullah, Mapping the city scale anthropogenic heat emissions from buildings in Kuala Lumpur through a top-down and a bottom-up approach, *Sustain Cities Soc* 76 (2022), 103443, <https://doi.org/10.1016/j.scs.2021.103443>.
- [59] J. Yan, S.H. Zhang, J.J. Zhang, S.X. Zhang, C.Y. Zhang, H. Yang, R.Z.Y. Wang, L. Y. Wei, Stormwater retention performance of green roofs with various configurations in different climatic zones, *J Environ Manag* 319 (2022), 115447, <https://doi.org/10.1016/j.jenvman.2022.115447>.
- [60] W. Yang, Y.Y. Wang, J.P. Liu, Optimization of the thermal conductivity test for building insulation materials under multifactor impact, *Constr Build Mater* 332 (2022), 127380, <https://doi.org/10.1016/j.conbuildmat.2022.127380>.
- [61] H. Zhang, C.Y. Shang, Tang GH Measurement and identification of temperature-dependent thermal conductivity for thermal insulation materials under large temperature difference, *Int J Therm Sci* 171 (2022), 107261, <https://doi.org/10.1016/j.ijthermalsci.2021.107261>.
- [62] W. Zhang, H. Min, X. Gu, Y. Xi, Y. Xing, Mesoscale model for thermal conductivity of concrete, *Constr Build Mater* 98 (2015) 8–16, <https://doi.org/10.1016/j.conbuildmat.2015.08.106>.
- [63] M.J. Zhao, P.C. Tabares-Velasco, J. Srebric, S. Komarneni, R. Berghage, Effects of plant and substrate selection on thermal performance of green roofs during the summer, *Build Environ* 78 (2014) 199–211, <https://doi.org/10.1016/j.buildenv.2014.02.011>.
- [64] X. Zheng, Z.J. Yang, J.C. Yang, M. Tang, C. Feng, An experimental study on the thermal and energy performance of self-sustaining green roofs under severe drought conditions in summer, *Energy Build* 261 (2022), 111953, <https://doi.org/10.1016/j.enbuild.2022.111953>.
- [65] W.K. Zhu, H. Lin, Y. Cao, B. Li, Thermal properties measurement of cut tobacco based on TPS method and thermal conductivity model, *J Therm Anal Calorim* 116 (2014) 1117–1123, <https://doi.org/10.1007/s10973-013-3543-z>.
- [66] Q.R. Zhuang, Z.M. Lu, Optimization of roof greening spatial planning to cool down the summer of the city, *Sustain Cities Soc* 74 (2021), 103221, <https://doi.org/10.1016/j.scs.2021.103221>.